

PROJECT REPORT

Phishing Website Detection Using Machine Learning Techniques

1. Objective

The objective of this project is to detect phishing websites using machine learning techniques. The system is trained on a dataset containing both phishing and legitimate URLs, and it extracts important URL-based features to classify whether a website is phishing or safe.

2. Software / Tools Used

- Google Colab
- GitHub
- Mendeley Desktop
- Python libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost

3. Dataset Description

The dataset used in this project is a phishing URL dataset containing extracted features from URLs. It includes both legitimate and phishing URLs. The dataset consists of multiple numerical and categorical features derived from URL structure and behavior.[1]

4. Methodology

4.1 Data Loading

The dataset was loaded into Google Colab using Pandas and stored in a dataframe for processing.

4.2 Data Preprocessing

- Removed irrelevant columns (e.g., Domain)
- Checked and handled missing values
- Shuffled dataset to avoid bias
- Split dataset into training and testing sets (80:20 ratio)

4.3 Feature Selection

Features related to URL structure were used for training. The target variable indicates whether a URL is phishing (1) or legitimate (0).[2]

5. Machine Learning Models Used

The following supervised machine learning models were implemented:

- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier
- Support Vector Machine (SVM)

Each model was trained on the dataset and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

6. Model Evaluation

6.1 Decision Tree

- Used tree-based classification
- Evaluated using accuracy and confusion matrix

6.2 Random Forest

- Ensemble of multiple decision trees
- Improved stability and accuracy

6.3 XGBoost

- Gradient boosting technique
- Provided highest accuracy among all models [3]

6.4 SVM

- Linear kernel-based classifier
- Used for binary classification

7. Results

All models were compared based on test accuracy. XGBoost performed the best among all models and was selected as the final model for prediction.

8. Model Saving

The final trained XGBoost model was saved for future use using serialization techniques.

9. Conclusion

This project successfully demonstrates the use of machine learning techniques for phishing website detection. Among all models tested, XGBoost provided the best performance. The system can help in identifying malicious websites and improving online security.

10. Future Scope

- Integration with real-time browser extension
- Use of deep learning models for better accuracy
- Expansion of dataset for improved generalization

11. References/Bibliography

- [1] S. Abu-Nimeh, D. Nappa, X. Wang, and S. Nair, "A Comparison of Machine Learning Techniques for Phishing Detection."
- [2] A. K. Jain and B. B. Gupta, "Phishing detection: Analysis of visual similarity based approaches," 2017, *Hindawi Limited*. doi: 10.1155/2017/5421046.
- [3] A. A. Orunsolu, A. S. Sodiya, and A. T. Akinwale, "A predictive model for phishing detection," *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 2, pp. 232–247, Feb. 2022, doi: 10.1016/j.jksuci.2019.12.005.